

$$L(\alpha, \beta) = \prod_{i=1}^n \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)}$$

$$\log L(\alpha, \beta) = \sum_{i=1}^n \log \frac{y_i^{\alpha-1} (1 - y_i)^{\beta-1}}{B(\alpha, \beta)}$$

$$= \sum_{i=1}^n [\log y_i^{\alpha-1} + \log(1 - y_i)^{\beta-1} - \log B(\alpha, \beta)]$$

$$= \sum_{i=1}^n [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i) - \log B(\alpha, \beta)]$$

$$= \sum_{i=1}^n [(\alpha - 1) \log y_i + (\beta - 1) \log(1 - y_i)] - n \log B(\alpha, \beta)$$

$$\log L(\alpha, \beta) = (\alpha - 1) \sum_{i=1}^n \log y_i + (\beta - 1) \sum_{i=1}^n \log(1 - y_i) - n \log B(\alpha, \beta)$$